



NQRUST-STORAGE

Global Open Technologies | Sovereign Indonesian Engineering

Enterprise storage platform with advanced storage management and unified multi-protocol access — managed from a single web console. Engineered in Indonesia.

4-in-1

S3+NFS+SMB
+iSCSI

CoW

Self-
Healing

Web

Console
Managed

RAID-Z

Data
Protection

Version 1.0 – Brochure
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Introduction

What Is NQRust-Storage?

A sovereign enterprise storage platform built on globally proven technologies — delivering advanced storage management with unified S3, NFS, SMB, and iSCSI protocol access.

NQRust-Storage is a sovereign enterprise storage platform engineered in Indonesia, built upon globally proven storage technologies. At its core, the platform is powered by an advanced **storage management engine** with a copy-on-write architecture, end-to-end checksums on every block, self-healing data repair, software-defined RAID, instant snapshots, inline compression, and native encryption. Storage is organized into **pools** (physical disk groups with configurable redundancy) and **datasets** (flexible logical filesystems with per-dataset properties, quotas, compression ratios, and mountpoints).

Built on top of this storage foundation, the platform delivers **unified block, file, and object storage** managed from a single web console — with built-in S3-compatible object storage, NFS and SMB file sharing, and iSCSI block export. The **NQRust Storage Console** unifies capacity management, service endpoints, and access governance in one operational console. A **Protocol Command Center** provides unified health and readiness monitoring for all four protocols in one dashboard view.

Storage Management — The Foundation

**Pools & Data Protection**

Create storage pools from physical disks with configurable redundancy: Mirror, RAID-Z1/Z2/Z3, or dRAID. No hardware RAID cards required. Expand pools by adding devices without disruption. Health, capacity, and error monitoring from the web console.

**Datasets & Properties**

Hierarchical datasets with per-dataset mountpoint, compression, quota, reservation, encryption, and readonly properties. Child datasets inherit from parents, overridable at any level. Writable clones from snapshots at zero cost. Block volumes (zvols) for iSCSI. Compression ratios visible per dataset.

**Integrity & Self-Healing**

Copy-on-write ensures data is not overwritten in place. End-to-end checksums detect silent corruption. Self-healing repairs data from redundant copies. Scheduled scrubs verify pool integrity. SSD read cache and write log for performance.

Protocol Services — Built on the Foundation

Four protocols serving data from pools and datasets

Protocol	Description	Security
S3 Object	Built-in S3-compatible engine. Buckets, access keys, storage limits, backup points, service logs, S3 browser.	Per-bucket key scoping, storage limits
NFS	Dataset-backed exports for Linux/macOS clients. Active session monitoring, quota coverage, export config.	Client IP/CIDR rules, root squash, sync/async
SMB	Dataset-backed shares for Windows/macOS/Linux. Users & groups, session tracking, network boundary, firewall.	Username/password auth, firewall rules
iSCSI	Block volumes exported to trusted initiators. I/O stats, volume resize, backup points, session control.	Initiator authentication, ACLs, CHAP

Platform Capabilities

Features That Protect Your Data

From storage management fundamentals to multi-protocol access and fleet operations.



Storage Pools

Create, import, scrub, and export pools. Mirror, RAID-Z1/Z2/Z3, dRAID protection. Real-time health, capacity, fragmentation, and dedup monitoring.



Datasets & Volumes

Hierarchical datasets with compression, quotas, reservations, encryption, mountpoints, readonly, and access controls. Block volumes (zvols) for iSCSI.



Snapshots & Clones

Fast copy-on-write snapshots. Compare side-by-side. Writable clones from any snapshot — instant, zero initial space. Sanoid automated scheduling with retention.



Replication & Syncoid

Native send/receive and Syncoid for production replication. Incremental sends, scheduled jobs, common snapshot tracking, and error handling.



Inline Compression

Transparent LZ4 and ZSTD compression at the storage level. Workload-dependent compression ratios visible per dataset in the console.



Encryption

Native dataset-level encryption. Data encrypted at rest. Per-dataset encryption keys. Integrates with snapshot and replication workflows.



Built-in S3 Engine

Local S3-compatible endpoint. Access keys with per-bucket scoping, storage limits, backup points, S3 browser, and full service logs.



Protocol Command Center

Unified dashboard showing health, readiness scores, sessions, capacity, and warnings for S3, NFS, SMB, and iSCSI in one view.



SSD Cache & Tiering

SSD read cache, SSD write log for sync acceleration. Dedicated metadata vdev on SSD for faster small-file I/O. Live hit rate, per-VDEV stats, auto-refresh dashboard.



SMART & Health Analysis

Disk health monitoring via SMART attributes, diagnostics, temperature. Pool health and scrub status analysis with historical tracking.



Observability & Logs

Pool history, pool events, cache statistics, searchable logs. Kernel debug log, system log, module parameters. Download all logs as ZIP for audit.



Fleet View

Monitor remote storage servers from one console via SSH. View pool status across your fleet. Centralized visibility for distributed deployments.



Storage Management Deep Dive

Pools built from vdevs with configurable redundancy. Special vdevs on SSD for metadata acceleration. Datasets with independent properties: compression, quotas, reservations, recordsize, encryption, readonly, atime, exec, multiple copies. Clones from snapshots at zero cost. Zvols for iSCSI and VM disks.



Built-in Admin Tools

Web Shell, Text Editor with syntax highlighting, File Browser, System Services viewer, SSH Connection manager, Network Bonding with rollback-safe failover, and NQRust Settings.

RAID & Protection Levels

Level	Description	Best For
Mirror	Full copy on each disk. Survives N-1 disk failures in the vdev.	Small pools, max performance
RAID-Z1	Single parity distributed across disks. Survives 1 disk failure per vdev.	General workloads
RAID-Z2	Double parity. Survives 2 disk failures per vdev.	Production recommended
RAID-Z3	Triple parity. Survives 3 disk failures per vdev.	Mission-critical, large arrays
dRAID	Distributed RAID with faster rebuild spread across all drives.	Large pools, faster resilver

Use Cases

Built for **Every** Data Challenge **Enterprise Private Cloud Storage**

Unified storage management + multi-protocol access for data centers. Pools and datasets organize capacity with per-dataset policies. NFS/SMB for file sharing, iSCSI for VM datastores, S3 for applications. **One console manages everything.**

 Government & Sovereign Data

Strong data sovereignty on your infrastructure, engineered in Indonesia. Self-healing integrity protects against silent corruption. Air-gapped deployment with no external dependencies. SMART monitoring and scrub scheduling for proactive maintenance.

 Backup & Disaster Recovery

Fast snapshots with Sanoid automated scheduling. **Syncoid replication** for production DR with incremental sends and operational error handling. S3-compatible backup target for Veeam, Restic, and Rclone. iSCSI backup points for block volumes.

 Multi-Site & Fleet Management

Fleet View monitors multiple remote storage servers from one console via SSH. Replication jobs sync data across sites. Network bonding supports storage traffic redundancy. Scale from a single server to multi-server and multi-site deployments.

Under The Hood

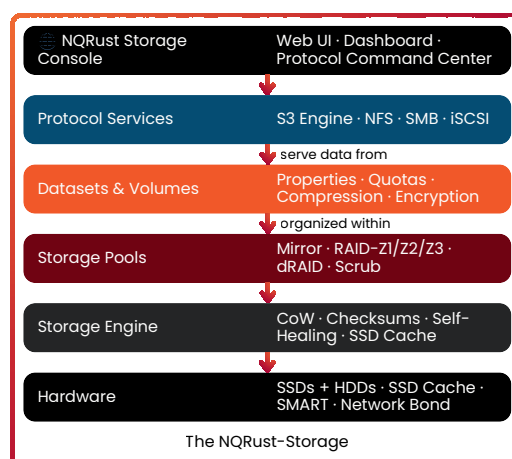
Storage Management Engine → Protocol Services → Operations & Fleet

The **storage engine** combines volume management and filesystem into one layer. Pools group physical disks into vdevs with configurable redundancy. Copy-on-write checksums every block, and self-healing repairs corruption from redundant copies.

Datasets are flexible filesystems on top of pools with independent properties: compression, quotas, encryption, recordsize. **Zvols** provide block devices for iSCSI and VMs.

Protocol services sit on datasets: S3 engine for objects, NFS and SMB for file shares, iSCSI for block volumes. Protocol Command Center monitors all four.

Operations: Sanoid for snapshots, Syncoid for replication, SMART for disk health, Fleet View for multi-server visibility.



Why NQRust-Storage for Your Organization?

Built on proven global technologies. Sovereign Indonesian engineering. Enterprise-ready from day one.

 Sovereign & Indonesian-Engineered

Built on globally proven storage technologies, engineered by Indonesian experts. All data stays on your infrastructure. Strong data sovereignty for government, defense, and regulated industries.

 Self-Healing Data Integrity

End-to-end checksums and automatic repair protect against silent data corruption. Copy-on-write ensures data is never partially written. Scrub scheduling for continuous verification.

 One Platform, Every Protocol

Pools, datasets, snapshots, replication, compression, encryption, and quotas as the foundation. Protocol services (S3, NFS, SMB, iSCSI) built on top of properly managed storage.

 Production-Ready Automation

Sanoid for automated snapshots. Syncoid for reliable replication. SMART monitoring for disk health. Fleet View for multi-server operations. All from one web console.